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The severity of earthquake events – statistical analysis and classification

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ABSTRACT

Earthquake events are natural disasters that can pose a threat to people's safety as well as their homes and possessions. In this paper, the severity level of earthquake disasters is addressed using the US National Oceanic and Atmospheric Administration (NOAA) database. A total of 5841 earthquake incidents are recorded that happened between 2150 BCE and 2015 CE. Few studies have done a comprehensive statistical analysis of the consequences of earthquakes. To address this gap, and after determining the probability distribution function of the number of fatalities, we evaluate the distribution of earthquakes with extreme fatalities to determine the severity levels according to the fatality-based disaster scale introduced by Wirasinghe, Caldera, Durage, and Ruwanpura [(2013). *Preliminary analysis and classification of natural disasters*. Proceedings of the ninth annual conference of the International Institute for Infrastructure, Renewal and Reconstruction (IIIRR), Queensland University of Technology, Brisbane, Australia, July 2013, Section B1.2, p. 11]. To this end, three different methods of determining the extreme events are considered: peak over threshold, *R*th order, and event-based and location-based block maxima. Moreover, a comprehensive collinearity analysis is performed to investigate any correlation and linear dependency between the earthquake parameters (magnitude, intensity, and focal depth) and the consequences in terms of earthquake fatalities. The severity classification based on block maxima has more detailed severity classes; hence, it is superior to the other two methods. For block maxima, the probability of a lower level disaster (Emergency to Catastrophe Type 1) being the extreme disaster is higher for the location (country)-based data set compared to the event-based worldwide data set, while the probability of a higher level disaster (Catastrophe Type 2 and above) being the extreme disaster is lower. These probabilities are to be expected because a single country, even over the full time period, is less likely to have a massive disaster compared to the world when a large number of extreme events, in this case 100, are considered.

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1. Introduction

Natural hazards are usually defined as extreme natural events that pose a threat to people and/or their property and possessions. Natural hazards that are geophysical in nature can be categorized as follows: geological (earthquakes, volcanoes, and landslides), atmospheric (windstorms, severe precipitation, temperature extremes, and lightening), and hydrological (floods and debris flows) (McGuire, Mason, Kilburn, Kilburn, & Kilburn, 2002). An earthquake is one of the main natural hazards with large ones being capable of triggering thousands of deaths and injuries, and causing significant property damage. Some earthquakes even trigger secondary natural disasters such as tsunamis, landslides, avalanches, and fires. The earliest written records of earthquakes are from 3000 years ago in China and 1600 years ago in Japan and the surrounding region. At present, millions of people and billions of dollars in terms of properties and public infrastructures are at risk due to potential earthquakes (Kramer, 1996).

An earthquake is a consequence of a sudden energy release in the earth's crust. It creates seismic waves with destructive power that cause buildings to collapse. Every earthquake has five different measures:

- Epicentre: the latitude and longitude of the point where an earthquake originates. In historical events, it may be referred to as the location of maximum ground motion.
- Focal depth: the depth below the earth's surface where the first motion of the earthquake happens.
- Magnitude: A measure of the size of the event, which refers to the amount of energy released during an earthquake.
- Origin time: the time when the earthquake started.

The destructive power of an earthquake is usually referred to as an earthquake's intensity. Various intensity scales describe an earthquake's magnitude. One of the first earthquake intensity scales was proposed by Rossi and Forel (1881) and was widely used in the late nineteenth century. The Rossi-Forel scale introduced 10 different intensity levels based on the level felt by observers and also the effects on buildings and natural features. In 1902, the Rossi-Forel scale was revised by Mercalli (1902) and was introduced as Mercalli Intensity Scale with the same number of levels; it formed the foundation of the first modern twelve degree scale known as Mercalli–Cancani–Sieberg (MCS) scale (Forel, 1881). The English version of MCS scale was published in 1931 by Wood and Frank Neumann as the Modified Mercalli Intensity (MMI). In this scale, the lower numbers indicated the intensity felt by the people and higher numbers were based on the degree of structural damage (Musson & Cecić, 2002). Other widely used scales, such as Richter's MM56 scale (1956), the Medvedev–Sponheuer–Karnik scale (1964), and the European Macro-seismic Scale, were developed by considering more quantitative aspects (Wood & Neumann, 1931). The common feature of the above-mentioned scales is the use of information obtained after the occurrence of the seismic event, via questionnaire surveys and field investigations, to determine the earthquake intensity level. Another significant approach to determine a scale of earthquake intensity is to calculate the original energy of shakes without considering the effects and consequences; this approach was developed by Richter in 1935 and is known as the Richter's Magnitude scale (Richter, 1935).

The third approach to classify the severity level of earthquakes, which has been studied less when compared to other approaches, is to utilize statistical analyses of the recorded data. Most of the statistical analyses of historical earthquake data mainly focused on seismic hazard evaluation and prediction. For instance, in an early attempt in Japan, a cyclic Poisson process and a general version of the self-exciting process were used to interpret the historical records (Vere-Jones & Ozaki, 1982). A comprehensive review of the probabilistic seismic hazard analysis (PSHA) indicated that this method provided an integrated analysis of all possible ground motions, considering historical data, at a specific site to represent the spectral domain and earthquake at that area. This method has been used as a basis for the seismic design of structures; however, it does not provide a scale or classification of earthquake intensity (McGuire, 2008).

The above methods address the intensity of a hazard. However, the impact of earthquakes on people, facilities, and the economy has not been studied in detail. Gad-el-Hak introduced a classification of disaster severity termed Disaster Scope, which has five levels according to the number of displaced/ tormented/ injured/ killed people or the affected area of the event (Gad-El-Hak, 2009), as shown in Table 1. The ranges proposed for casualties and the area affected appear to be arbitrary (Caldera & Wirasinghe, 2014). Wirasinghe et al. (2013) developed a classification method for natural disaster impacts in general based on the statistical analysis of fatality data and introduced a set of modified and accurately defined terminologies to define different severity levels. Following the same approach, Caldera and Wirasinghe (2014) classified the severity level of volcanic eruptions considering historical data of several parameters such as the number of fatalities, injuries, and houses damaged. While these two studies mainly focused on natural disasters in general and volcanic eruptions, no studies have been conducted to classify the severity level of earthquakes. This lack of classification for earthquakes necessitates developing a fatality-based severity level considering extreme seismic events.

While most of the previous statistical studies have focused on determining the frequency and probability distribution of foreshocks and aftershocks (Ogata, 1981; Omori, 1894; Utsu, 1961; Utsu & Ogata, 1997), forecasting the occurrence of the earthquakes (Kagan & Knopoff, 1981; Murru, Console, & Falcone, 2009; Tiampo & Shcherbakov, 2012; Uyeda, Nagao, & Kamogawa, 2009) or a secondary earthquake's impact (landslide (Keefer, 2000; Qi, Xu, Lan, Zhang, & Liu, 2010) and rock fall (Marzorati, Luzi, & De Amicis, 2002)), very few of them performed statistical analysis for the consequences of earthquake. Considering the damage caused by an earthquake, Rowshandel et al. (2006) estimated the expected future earthquake economic loss in the San Francisco Bay Area

Table 1. Disaster scope (Gad-El-Hak, 2009).

Scope		Number of casualties		Geographic area affected
I	Small disaster	<10	or	<1 km^2
II	Medium disaster	10–100	or	1–10 km^2
III	Large disaster	100–1000	or	10–100 km^2
IV	Enormous disaster	1000–10,000	or	100–1000 km^2
V	Gargantuan disaster	>10,000	or	>1000 km^2

based on the historical earthquake data. They estimated the overall long-term damage and loss using the PSHA maps for the State of California. Taking fatality into consideration, Tsai, Yu, Chao, and Lee (2001) performed statistical analysis for the fatality data of the Chi-Chi, Taiwan earthquake in terms of spatial distribution and age dependency. They also analysed the empirical time functions of the cumulative number of fatalities, injuries, and missing people to show how critical the search-and-rescue operations were in the first 48 hours of the Kobe, Japan and Chi-Chi, Taiwan earthquakes. Considering the same earthquake consequences, Coburn, Pomonis, and Sakai (1989) sorted the earthquake fatalities by cause. They obtained a cumulative probability distribution function (PDF) of both collapsed buildings and mortality of population for different building types using a proprietary data set. Spence (2007) compared the earthquake death rate over the past 50 years for high, medium, and low human development indices (HDI). Their analysis showed that countries with high HDI such as USA and Japan were successful in decreasing the earthquake death rate over the past 50 years. However, there is no significant reduction in the earthquake death rate for countries with a low or medium HDI, such as Iran. Moreover, they analysed the cause of the injury, including structurally and non-structurally induced injuries, for different severity levels in some deadly earthquakes over the past 50 years.

Although all of these studies have focused mainly on mitigating the number of fatalities and economic loss, none have dealt with the PDF of these consequences or estimated these consequences based on other earthquake parameters. Nichols and Beavers (2003) showed how to develop a simple fatality function to estimate human losses. They used regression analysis to determine losses for a specific area struck by a specific earthquake. They also used some multiplicative factors, including population densities, total population, distance from the epicentre to the population, and building and ground-type differences to increase the accuracy of the estimation made by the regression model. Nichols and Beavers (2008) performed a statistical analysis of the world's historically recorded earthquake fatality data set. They reviewed and used the data from the NOAA to establish PDFs of the fatalities for three different time periods: 186 B.C.E.–1999 C.E., 186 B.C.E.–1899 C.E., and 1900–1999 C.E. Linear regression analysis was performed to estimate the count of fatal earthquakes in a given year. While these two latter studies tried to estimate fatalities caused by the earthquake, they did not determine the PDF of fatalities. Further, none of the previous studies have considered the distribution of the historical extreme fatality events, which is essential to determine the severity level of these events. To address these issues, after determining the PDF of historical earthquake fatalities, we investigate the PDF of extreme events. Finally, considering the PDF of extreme events, a fatality-based severity level can be introduced.

The paper is organized as follows: Section 2 presents the basic statistical analysis conducted for the number of fatalities during an earthquake. We discuss in particular which PDF best fits the historical events. In Section 3, linear dependency between earthquake parameters and fatalities is investigated. In Section 4, the pdf of fatalities during extreme events is determined using three approaches; these pdfs are then used to determine a severity scale for extreme events. Section 5 summarizes the applied fatality-based severity level. The results are discussed in Section 6. Finally, conclusions are provided in Section 7.

2. Descriptive analysis of parameters of earthquake disasters

The NOAA database provides detailed information on historical earthquake events that happened between 2150 B.C.E. to the present. For each specific event, it includes the date, location (name, latitude, and longitude), earthquake parameters (focal depth, magnitude, and intensity), and earthquake effects (deaths, injuries, damages, houses destroyed, and houses damaged). For each event, the moment magnitude is expressed as a number in the range of 0.0–9.9 and is related to the total energy released at the epicentre. Because the magnitude of an earthquake is not a good criterion to estimate the total physical damage caused, the intensity of each event needs to be known. The intensity in the NOAA database is expressed as the MMI given in Roman numerals ranging from 1 to 12. This intensity refers to the maximum reported intensity (NOAA, 2015). Table 2 shows the conversion method used by the NOAA database to convert the other intensity scales to the MMI scale.

The NOAA database provides a variety of earthquake impacts such as fatalities, injuries, and damages. However, only the number of fatalities is considered for the analysis of this study. The number of deaths due to tsunami and volcanic eruptions caused by earthquakes are excluded from the analysis.

Before starting the analysis, it is essential to prove the stationarity of the fatality data used in this study. Figure 1 shows the plotted fatality data along with the fatality local average versus time. The blue dots shows the actual fatalities associated with each historical event and the red lines show the average fatalities in each time period of 50 years. The time range, 1150 C.E.–2015 C.E., is divided into 16 fifty-year periods with the last period being 65 years. Because only 65 records are available for the earthquake events that happened between 2150 B.C.E. and 1150 C.E., and 66 historical divisions exist in this range, the events in this time period are excluded from the stationarity analysis.

Figure 1 exhibits a considerable temporal structure with trends and cyclic swings at various frequencies. Based on the research of Koutsoyiannis (2011), distinguishing stationarity from non-stationarity is a matter of answering the question: does the red line in the above figure represent a known (deterministic) function or unknown (random) function? If the causal mechanisms responsible for all transitions is perfectly understood, then a non-stationary description can be adopted. If, on the other hand, it is uncertain why the observed variations occurred, then a stationary description should be used. In other words, a stationary time series will have no predictable patterns in the long term. In

Table 2. Comparison of Modified Mercalli (MM) and other intensity scales (NOAA, 2015).

Modified Mercalli	Rossi-Forel	Japanese	European
I	I	0	I
II	I–II	I	II
III	III	II	III
IV	IV–V	II–III	IV
V	V–VI	III	V
VI	VI–VII	IV	VI
VII	VIII–VIII+ to IX–	V	VIII
IX	IX+	V–VI	IX
X	X	VI	X
XI	–	VII	XI
XII	–	–	XII

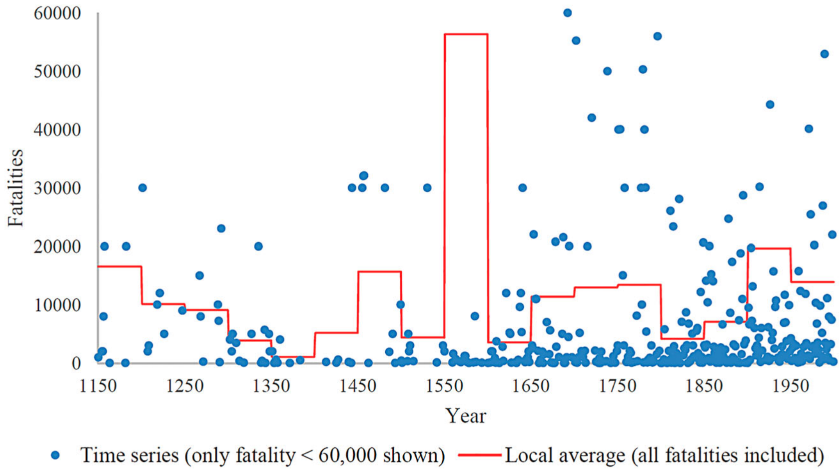

Figure 1. Historical fatalities with the local average for 50 years intervals.

Figure 1, data stationarity is assumed in this case because the function of cyclic swings is unknown and no predictable pattern is observed.

Table 3 shows the descriptive statistical analysis for fatality performed using IBM SPSS. From the total of 5841 earthquake events recorded in the NOAA database, data on the number of deaths are available for 1882 events. The most observed value is 1. The maximum value is 830,000. These values indicate that data are dispersed over a wide range. The variance of this parameter is great and does not provide useful information regarding the dispersion of data. In this case, the coefficient of variation (c.o.v.) is a better index. The c.o.v. is very high for this parameter, which shows that the data are highly dispersed around the mean.

The high and positive value of skewness for this parameter shows that the data above the mean are widely dispersed. The median is significantly larger than the mean, which confirms the existence of a long tail on the right side of the mean value. The high values of kurtosis for the death parameter indicate that the distribution tends to have a peak near the mean, which is accompanied by heavy tails.

Table 3. Basic statistics for number of deaths caused by earthquakes.

		Deaths
N	Valid	1882
	Missing	3959
Mean		4072
Median		29
Mode		1
Std. deviation		26,466
Variance		700,483,480
Coefficient of variation		6.50
Skewness		19.57
Kurtosis		533.08
Minimum		1
Percentiles	25	4
	50	29
	75	400
Maximum		830,000

2.1. Fitting distribution into parameters that reflect the severity level

The next step is to find the distribution which best fits the data using the Anderson–Darling (A–D) statistical test and the graphical tools, P–P plot and Q–Q plot, using Minitab software. The A–D test measures how well the data fit a specific distribution. The null hypothesis is that data follow a specific distribution and the p-value shows if the null hypothesis is correct or not. The smaller values of A–D suggest a better distribution for specific data. The reason for choosing A–D over other hypothesis tests, such as Kolmogorov–Smirnov (K–S) or Shapiro–Wilk (S–W), is that A–D is more sensitive near the tails of the distribution in comparison to other hypothesis tests and, hence, will behave more powerfully in the case of our data, which are dispersed over a long range.

The death parameter is found to be distributed as a two-parameter Weibull Distribution ($k = 0.30855$, $\lambda = 252.74817$) at a significance level of $\alpha = 0.005$. The related PDF is

$$f(x) = 0.056(x)^{-0.69145} \exp\left[-\left(\frac{x}{252.74817}\right)^{0.30855}\right]. \quad (1)$$

3. Correlation analysis

Correlation analysis is performed to discover any linear dependency between fatalities and earthquake parameters, including magnitude, intensity, and focal depth. The output of the correlation analysis for two different variables is the Pearson product-moment correlation coefficient or, simply, a correlation coefficient, which is a measure of the linear correlation between two random variables (Zholud, 2011). A correlation coefficient accepts a value between −1 and 1. The linear dependency between two variables is greater the closer the absolute correlation coefficient is to 1. The results of the correlation analysis are shown in Table 4.

In general, for the same population density and level of structural integrity, a higher number of fatalities occur during earthquakes with a higher magnitude or intensity. Therefore, determining whether or not fatality has a linear dependency on these parameters is important.

While the positive sign of the correlation coefficient (0.171 for death magnitude and 0.236 for death intensity) shows that increasing the magnitude or intensity will increase fatalities, the low value of the correlation coefficients (lower than 0.5) indicates that there is not enough evidence to confirm a linear relationship between number of deaths and these parameters. The intensity parameter has a greater correlation coefficient than the magnitude parameter. This difference is mainly because the intensity scale is based on the degree of structural damage and not on the amount of energy released by an earthquake. Intensity has a greater correlation with fatalities because collapsing buildings tend to be the most deadly factor during earthquakes. For the most recent earthquakes, the

Table 4. Pearson product-moment correlation coefficient matrix.

	Intensity (MMI)	Magnitude	Focal depth
Death	0.236	0.171	−0.068

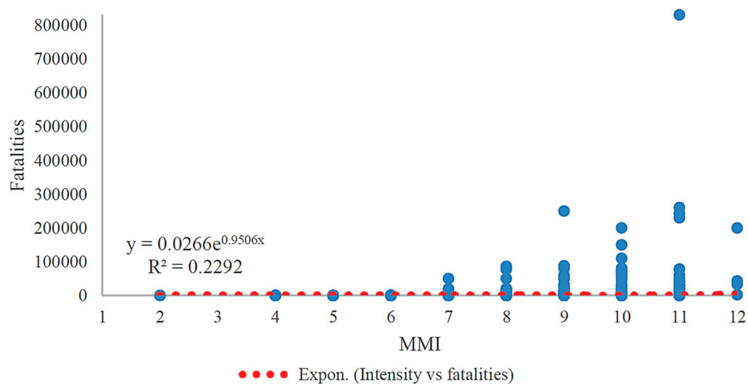

Figure 2. Fatalities versus intensity.

correlation coefficient between intensity and number of deaths has been further studied. Several historical time intervals were chosen such as events after 1900, events after 1950, and events after 1990. However, the correlation did not change significantly and 0.236 is still the highest correlation coefficient obtained for this database.

A non-linear relationship between death and intensity was also investigated using non-linear regression analysis. The results (Figure 2) showed that there is not a non-linear relationship between death and intensity because of the very low R^2 value. However, conducting ANOVA showed that the equation and both coefficients of the equation, shown in Figure 2, were significant. The results indicated that other factors related to proper mitigation, such as evacuation, early warning, and awareness, may lower the correlation between intensity and fatality.

In contrast, because the exact location of the earthquake is not traceable in the data set, the affected area is not obvious. Some earthquakes that release a lot of energy (high magnitude) may occur in low-density residential areas or regions, such as oceans, mountains, and deserts, and, therefore, are not destructive. Moreover, an earthquake with a lower Richter number (lower magnitude) may occur in a high-density population area and, consequently, become very destructive. For example, in 2003, an earthquake happened in an urban area with a high population density in Bam, Iran with a magnitude of 6.6 and killed 31,000 people. In contrast, there are numerous instances of a high moment magnitude earthquake with few or zero fatalities. The 2014 Iquique earthquake (with the moment magnitude of 8.2 and 7 fatalities) and the 2014 Alaska earthquake (with the moment magnitude of 7.9 and no fatalities) are two examples.

4. Extreme value analysis

Extreme value theory was proposed by Gumbel in 1958 to analyse extreme values in data sets (Reiss & Thomas, 2007). Three different approaches were suggested to determine the subset of extreme values:

- Block maxima (minima): In this approach, the main data set is divided into equal blocks and the maximum (minimum) value of each block is determined as a single member of the subset. The commonly used parameter to determine the blocks is 'yearly'. However, it can vary according to the nature of a specific data set.

- *R*th order: This approach is an improved form of the block maxima method where the first *r*th largest values in each block are determined, which increases the level of data utilization and reduces the bias in data according to the best selected *R* value.
- Peak over (below) threshold: In this method, the values which exceed (fall below) a certain threshold form the extreme value data set.

Six different distributions used to fit the extreme data include: Weibull, Frechet, and Gumbel in block maxima and *R*th order (Lehmann & Casella, 1998), and Exponential, Pareto, and Beta in peak over (below) threshold (Reiss & Thomas, 2007). All of the above methods are examined in this paper to determine extreme fatalities of earthquake events.

4.1. Block maxima

Two different approaches are taken into account to determine the blocks. In the first approach, the length of each block is considered equal to 100 distinctive events sorted by the time of occurrence. In the second method, each country is considered as a unique block.

4.1.1. Blocking based on event

Because the data set includes 5840 distinctive events based on the block length of 100 events, 59 blocks were produced. First, the maximum number of fatalities in each block was selected to evaluate the extreme value distribution. The parameters of these distributions, Weibull, Frechet, and Gumbel, need to be known to plot their cumulative distribution function (CDF) along with the empirical CDF. To this end, the data set of fatalities was first imported to EasyFit 5.5 Professional to determine the distribution parameters. Knowing the mathematical equation of these distributions, we plotted the CDF using IBM SPSS or Microsoft Excel. The CDFs of the extreme value distributions are shown in Table 5.

Table 5. CDF equations for extreme value distributions.

Distribution	CDF	Note
Two-parameter Frechet	$F(x) = \exp\left(-\left(\frac{\beta}{x}\right)^{\alpha}\right)$	α – continuous shape parameter ($\alpha > 0$) β– continuous scale parameter ($\beta > 0$) γ – continuous location parameter ($-\infty < \gamma < \infty$)
Three-parameter Frechet	$F(x) = \exp\left(-\left(\frac{\beta}{x-\gamma}\right)^{\alpha}\right)$	
Two-parameter Weibull	$F(x) = 1 - \exp\left(-\left(\frac{x}{\beta}\right)^{\alpha}\right)$	α – continuous shape parameter ($\alpha > 0$) β – continuous scale parameter ($\beta > 0$) γ – continuous location parameter ($-\infty < \gamma < \infty$)
Three-parameter Weibull	$F(x) = 1 - \exp\left(-\left(\frac{x-\gamma}{\beta}\right)^{\alpha}\right)$	
Gumbel max	$F(x) = \exp\left(-\exp\left(\frac{\mu - x}{\sigma}\right)\right)$	σ – continuous scale parameter ($\sigma > 0$) μ – continuous location parameter ($-\infty < \mu < \infty$)

Finally, the best extreme value distribution for the extreme value was chosen by comparing the empirical CDF of the five distribution functions in Table 5. For a precise comparison, the mean absolute error (MAE) of each extreme value distribution with regard to the empirical CDF was calculated using Equation (2) (Lehmann & Casella, 1998), where $|\widehat{Y}_i - Y_i|$ is the absolute error of the ith fatality number. The distribution with the minimum MAE is then chosen as the best fitted distribution. Figure 3 shows the CDF of different distributions plotted on the same graph along with the CDF of the extreme values.

$$\mathrm{MAE} = \frac{1}{n}\sum_{i=1}^{n} |\widehat{Y}_i - Y_i|. \tag{2}$$

For these distributions, the MAE values are compared, as shown in Table 6, to determine the best fitted CDF. The three-parameter Frechet offers the minimum MAE, which indicates that it is the best fitted distribution to the extreme events. The CDF of the three-parameter Frechet distribution is shown in Equation (3).

$$F(x) = \exp\left(-\left(\frac{21,639}{x + 5187.6}\right)^{1.0422}\right). \tag{3}$$

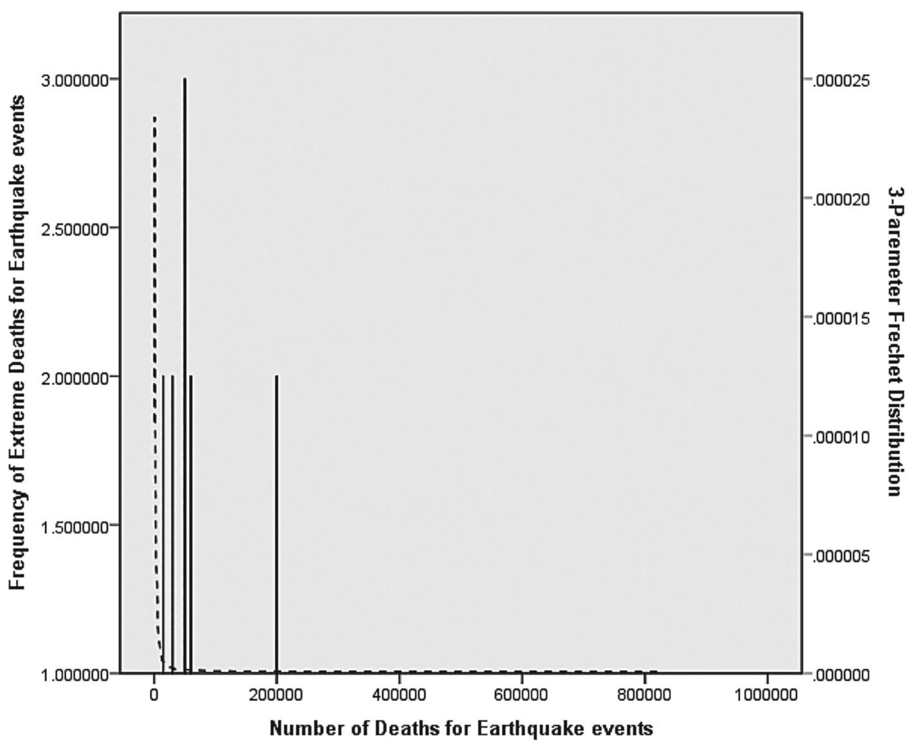

Figure 3. Histogram of extreme death for earthquake events and the fitted three-parameter Frechet distribution (block maxima event-based).

Table 6. MAE for extreme value distributions of death.

	Frechet (2P)	Frechet (3P)	Weibull (2P)	Weibull (3P)	Gumble
MAE	0.053	0.015	0.025	0.042	0.125
Rank	4	1	2	3	5

4.1.2. Blocking based on location

In this approach, each country is considered as a block, and the extreme events are chosen from the events that happened regardless of the time they occurred. Although choosing the block based on the occurrence time helps us to better analyse extreme events in regard to the gradual change in human civilization, we will face some challenges that may affect the accuracy of the analysis. First, the extreme values determined based on the event-based approach belong to a limited number of countries/ regions. As shown in Table 7, the 59 extreme events belong to only 21 countries out of 166 countries and regions in the data set, taking into consideration the number of fatalities.

More specifically, approximately half of the extreme events happened in China, Iran, and Turkey, which proves that the event-based approach tends to overemphasize the events in a few countries. However, the event-based approach tends to ignore the extreme events that happened in other earthquake-prone countries. For example, the United States of America (251 historical events), Greece (251 events), Philippines (202 events), Chile (186 events), and Taiwan (90 events) are not represented in Table 7. Thus, blocking based on events is biased towards the location. In addition, this approach tends to overemphasize the recent extreme events because there is less earthquake data available from 2150 B.C.E. to 1900 C.E. than in recent years. Figure 4 represents the dispersion of the data on deaths for the two different approaches for determining blocks.

In event-based approach, the extreme events are more concentrated around the past 150 years (67%) and very few extreme events belongs to the time period before 1500 C. E. However, blocking the data set based on the location provides extreme events that are more dispersed. Table 8 shows the extreme events frequency for different time intervals.

It is better to conduct the extreme value analysis by blocking based on the location because blocking based on the time of events is biased towards both time and location of the earthquake. In the new approach, each country is considered as a separate block.

Table 7. Extreme values frequency and total frequency for different countries.

Country	No. of extreme events	Total frequency	Country	No. of extreme events	Total frequency
China	10	576	Yemen	1	10
Iran	10	368	Turkmenistan	1	12
Turkey	7	326	Peru	1	178
Japan	6	389	Haiti	1	15
Pakistan	3	55	Mexico	1	196
India	3	102	Algeria	1	55
Italy	3	319	Portugal	1	23
Ecuador	3	64	Israel	1	26
Azerbaijan	2	15	Afghanistan	1	53
Indonesia	1	340	Nepal	1	13
Morocco	1	19			

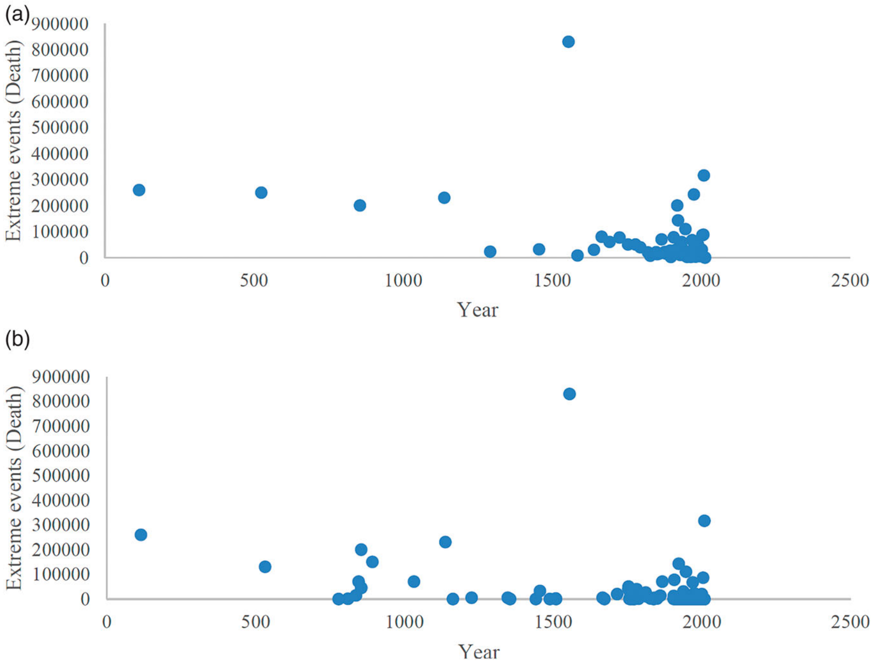

Figure 4. (a) Event-based approaches for choosing blocks. (b) Location-based approaches for choosing blocks.

Consequently, 118 unique blocks are created, and the extreme value analysis is performed for fatalities. The weakness of this approach is the disparity in the number of earthquake fault lines found in large and small countries.

While the MAE values for the two-parameter Weibull and three-parameter Weibull are close, Table 9 shows that the three-parameter Weibull has a lower MAE value; hence, it is better than the two-parameter Weibull. This result is mainly because of the superior fit of the three-parameter Weibull (Figure 5) to the empirical data in the lower part of the data set (for fatalities less than 2000). However, for fatalities greater than 2000, the two-parameter Weibull fits better than the three-parameter Weibull. The best fitted distributions in this case can be expressed as follows:

$$\begin{aligned} &\text{If } x \leq 2000 \qquad F(x) = 1 - \exp\left(-\left(\frac{x-1}{3439.6}\right)^{0.21862}\right), \qquad \text{three - parameter Weibull.} \\ &\text{If } x > 2000 \qquad F(x) = 1 - \exp\left(-\left(\frac{x}{2837.6}\right)^{0.30824}\right), \qquad \text{two - parameter Weibull.} \end{aligned} \tag{4}$$

Table 8. Extreme events frequency for different time intervals.

Time interval	2150 B.C.E.	0–500 C.E.	500 C.E.–1000 C.E.	1000 C.E.–1500 C.E.	1500 C.E.–2015 C.E.
Blocking by event	0%	2%	3%	5%	90%
Blocking by location	0%	1%	8%	8%	83%

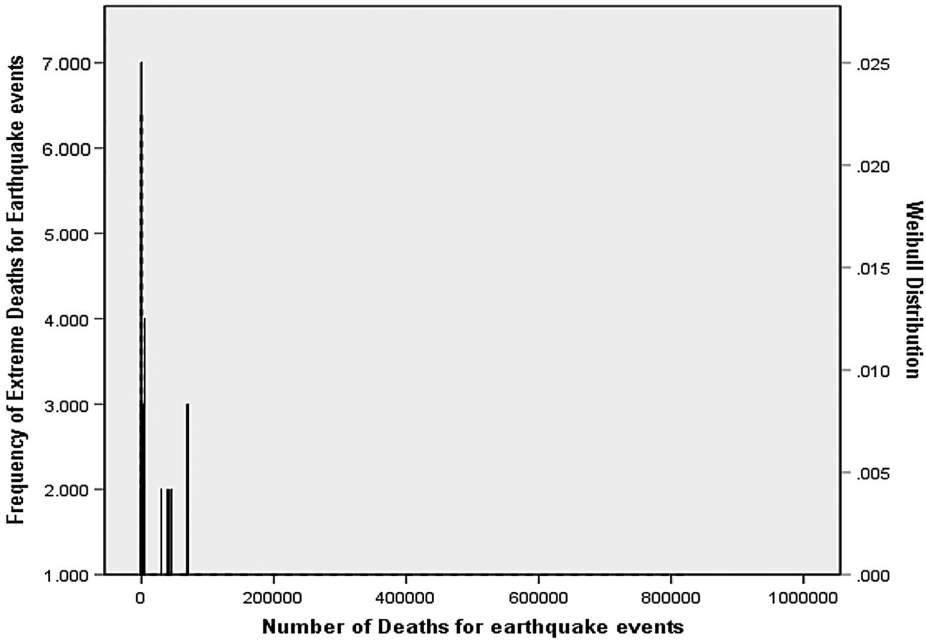

Figure 5. Histogram of extreme death for earthquake events and the fitted three-parameter Weibull distribution (block maxima-location-based).

Table 9. MAE for extreme value distribution of death.

	Frechet (2P)	Frechet (3P)	Weibull (2P)	Weibull (3P)	Gumble
MAE	0.054	0.172	0.043	0.024	0.148
Rank	3	5	2	1	4

4.2. Threshold method

The second approach for determining the extreme data set is the threshold method, which analyses the data that exceed a given threshold level u. The important issue is the choice of the threshold u. A high level of u results in few extreme events and high variance estimations. However, a low level of u gives a wide range of data, which may contain data that are not considered as extreme events. Boos (1984) recommends $k = 0.02n$ for $50 \leq n \leq 500$ and $k = 0.1n$ for $500 \leq n \leq 5000$, where n is the size of data set and k is the number of datum that exceed the threshold level u. In this regard, Hasofer (1996) suggested that k can be taken to be equal to $1.5\sqrt{n}$. We used the approach presented by Hasofer (1996) to find the threshold level. A total of 1882 data is available on the number of deaths; of this total, 65 exceed the threshold level u, which means that the threshold level u equals 30,000. To determine the severity level of earthquakes, we find the distribution that fits the data over the threshold. The peaks over the threshold follow a generalized Pareto distribution, which comes in three forms: Exponential, Pareto, and Beta. The PDF of the mentioned distributions is shown in the Table 10.

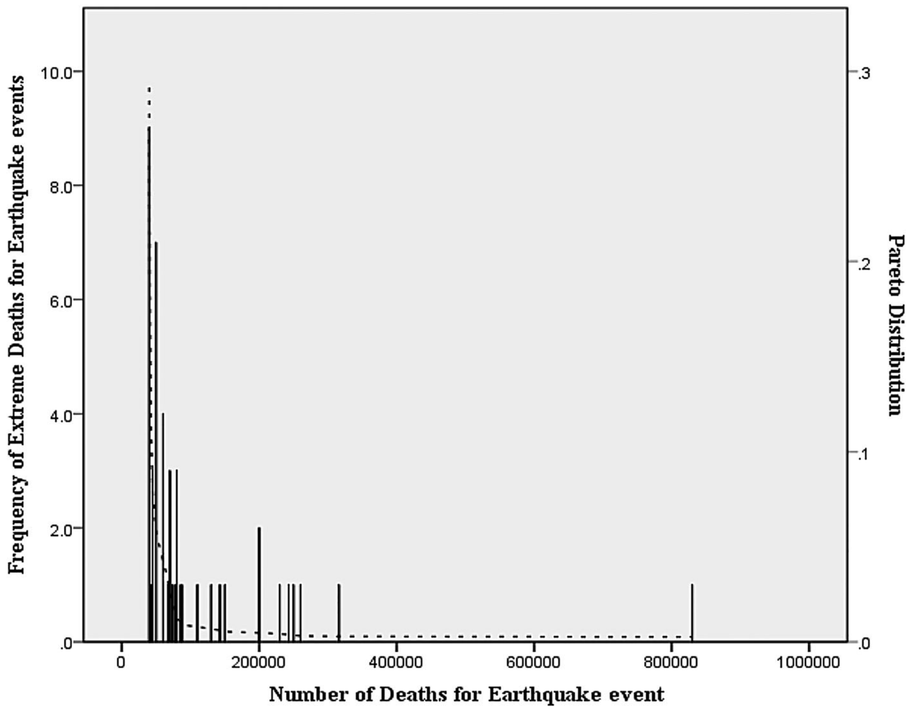

Figure 6. Histogram of extreme death for earthquake events and the fitted Pareto distribution (threshold).

By plotting the CDF associated with each of the distributions and comparing them with the empirical CDF, we determine the best distribution. To find the best fitted distribution for our data, we estimate the MAE values associated with each of the distributions in Table 11.

The estimated value of MAE for the Pareto distribution is 0.019, which is the lowest value among the four distributions. This result indicates that the Pareto distribution is the best fitted distribution, and Beta is a close second. The extreme death events are found to be distributed as a Pareto ($\alpha = 1.4086$, $\beta = 30{,}000$) distribution (Figure 6). The CDF of this distribution is represented in Equation (5).

$$F(x) = 1 - \left(\frac{30,000}{x}\right)^{1.4086}. \tag{5}$$

Table 10. PDF equations for extreme value distributions (threshold).

Distribution	PDF	Note
Exponential	$f(x) = \alpha \exp(-\alpha x)$	α – inverse scale parameter ($\alpha > 0$)
Two-parameter exponential	$f(x) = \alpha \exp(-\alpha(x-\gamma))$	γ – continuous location parameter ($x \geq \gamma$)
Pareto	$f(x) = \frac{\alpha\beta^{\alpha}}{x^{\alpha+1}}$	α – continuous shape parameter ($\alpha > 0$) β – continuous scale parameter ($\beta > 0$)
Beta	$f(x) = \frac{x^{\alpha-1}(1-x)^{\beta-1}}{B(\alpha, \beta)}$	α, β – continuous shape parameters ($\alpha, \beta > 0$)

Table 11. MAE for extreme value distributions (threshold) of death.

	Exponential	Exponential (2P)	Pareto	Beta
MAE	0.037	0.061	0.019	0.020
Rank	3	4	1	2

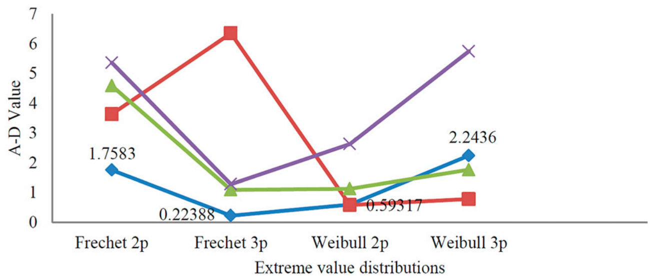

Figure 7. Anderson–Darling test results for block length = 100 events and different R values.

4.3. Rth order method

The third approach to determine the subset of extreme events is the *R*th order method, which starts with choosing the most appropriate block length and R value. For this purpose, we tried to define the blocks of data based on yearly time periods, which is a commonly used parameter. However, it was found that there are more data on events in recent history compared to earlier in history, which results in losing a large amount of data if the blocks are determined based on multi-year time periods. Therefore, in this method, the blocks are determined based on equal groups of recorded events.

Assigning an appropriate length (number of events) for each group is the next step to provide the extreme subset. Selection of block length is a compromise between bias and variance; a small length produces a large number of blocks and may cause an underestimated standard deviation, and a large group length gives few block averages for calculation (Lehmann & Casella, 1998). Choosing the value of R can also affect the statistical analysis in the same way. The main objective of this study is to determine a fatality-based severity scale based on the best extreme value distribution. To achieve this goal, we analysed different combinations of block length with R values based on the results of the Anderson–Darling test to determine the most appropriate value. To this end, different block lengths, including 100, 200, and 300, were considered and different R values were analysed to determine which combination of block length and R value produces the lowest A–D value. Figure 7 shows the A–D value associated with different combinations of block length (100) and R values (1, 2, 3, and 4). As illustrated in this figure, the best data set was a 100 event block length with $R = 1$ with the least value of A–D equal to 0.22388. Because this data set is exactly the same as the set produced using the block maxima method with 100 blocks, the best fitted distribution function is the same, which is the three-parameters Frechet ($\alpha = 1.0422$, $\beta = 21{,}639$, $\gamma = -5187$).

5. Severity classification

In this section, the severity level of earthquakes is determined based on fatality data of historical extreme earthquakes. According to a method proposed by Wirasinghe et al. (2013)

Table 12. Fatality-based disaster scale for general natural disasters (Caldera & Wirasinghe, 2014).

Type	Fatality range	Example
Emergency	$1 \leq F < 10$	A small landslide that kills one person
Disaster Type 1	$10 \leq F < 100$	Edmonton Tornado, Canada – 1987 that killed 27 people
Disaster Type 2	$100 \leq F < 1000$	Thailand flood – 2011 that resulted in a total of 815 deaths
Catastrophe Type 1	$1000 \leq F < 10{,}000$	Hurricane Katrina – 2005, USA that killed 1833 people
Catastrophe Type 2	$10{,}000 \leq F < 100{,}000$	Tohuku earthquake and tsunami – 2011, Japan that killed 15,882 people
Calamity Type 1	$100{,}000 \leq F < 1M$	Haiti earthquake 2010 killed 316,000 people
Calamity Type 2	$1M \leq F < 10M$	China floods – 1931 death toll >2,500,000
Cataclysm Type 1	$10M \leq F < 100M$	–
Cataclysm Type 2	$100M \leq F < 1B$	Super Volcano (e.g. Yellowstone): estimated deaths <1B
Partial or full extinction	$1B \leq F$	• Meteor strike (diameter >1.5 km) – estimated deaths: <1.5*109 • Pandemic (Avian influenza) – estimated deaths: <2.8B

for a unified classification of all types of natural disasters, and further developed for volcanic eruptions by Caldera and Wirasinghe (2014), a logarithmic scale has been used to classify severity levels as shown in Table 12. In this method, the extreme value PDF is divided into levels based on the mean value and standard deviation of the data. The lowest limit of the scale is the occurrence of one death and the scale increases based on a log scale for the higher levels. Earthquake severity levels are determined in the following subsections according to Frechet and Weibull distributions. Earthquake severity levels are not determined according to Pareto distribution as the extreme range starts from 30,000 deaths, which is high for the lowest extreme event and, it is unrealistic to neglect a large number of severe earthquakes with fatalities less than 30,000.

5.1. Severity classification based on block maxima method – event-based blocks

The severity thresholds can be determined by setting $\mu - k_i\sigma$ equal to the interval bounds shown because the mean value, 71,015.7, and standard deviation, 125,772.5, of the data set created by the block maxima method is known. Using the approximate mean value and standard deviation as 70,000 and 125,000, respectively, we determine k_i values for each severity level. The fatality-based severity levels, which range from 'Emergency' to 'Calamity Type 2 and higher', are represented in Table 13.

Table 13. Classification of fatality for event-based block maxima.

Type	Fatality range		Probability	Earthquake Example
Emergency	$1 \leq F < 10$	$\mu - 0.559992\sigma \leq F < \mu - 0.55992\sigma$	0.0001	Sistan, Iran – 1994 – killed 6 persons
Disaster Type 1	$10 \leq F < < 100$	$\mu - 0.55992\sigma \leq F < \mu - 0.5592\sigma$	0.0010	Tachilec, Myanmar – 2011 – killed 75 persons
Disaster Type 2	$100 \leq F < 1000$	$\mu - 0.5592\sigma \leq F < \mu - 0.552\sigma$	0.0121	Van, Turkey – 2011 – killed 589 persons
Catastrophe Type 1	$1000 \leq F < 10{,}000$	$\mu - 0.552\sigma \leq F < \mu - 0.48\sigma$	0.2104	Mexico City, Mexico – 1985 – killed 9500 persons
Catastrophe Type 2	$10{,}000 \leq F < 100{,}000$	$\mu - 0.48\sigma \leq F < \mu + 0.24\sigma$	0.5895	Pisco, Peru – 1970 – killed 66,794 persons
Calamity Type 1	$100{,}000 \leq F < 1M$	$\mu + 0.24\sigma \leq F < \mu + 7.44\sigma$	0.1569	Shaanxi, China, 1556 – killed 830,000 persons
Calamity Type 2 and higher	$1M \leq F$	$\mu + 7.44\sigma \leq F < \mu + 79.44\sigma$	0.0181	No historical records

Table 14. Classification of fatality for location-based block maxima.

Type	Fatality range		Probability	Earthquake example
Emergency	$1 \leq F < 10$	$\mu - 0.3111\sigma \leq F < \mu - 0.311\sigma$	0.2386	Sistan, Iran – 1994 – killed 6 persons
Disaster Type 1	$10 \leq F < 100$	$\mu - 0.311\sigma \leq F < \mu - 0.31\sigma$	0.1304	Tachilec, Myanmar – 2011 – killed 75 persons
Disaster Type 2	$100 \leq F < 1000$	$\mu - 0.31\sigma \leq F < \mu - 0.3\sigma$	0.1648	Van, Turkey – 2011 – killed 589 persons
Catastrophe Type 1	$1000 \leq F < 10{,}000$	$\mu - 0.3\sigma \leq F < \mu - 0.2\sigma$	0.2373	Mexico City, Mexico – 1985 – killed 9500 persons
Catastrophe Type 2	$10{,}000 \leq F < 100{,}000$	$\mu - 0.2\sigma \leq F < \mu + 0.8\sigma$	0.1790	Pisco, Peru – 1970 – killed 66,794 persons
Calamity Type 1	$100{,}000 \leq F < 1M$	$\mu + 0.8\sigma \leq F < \mu + 10.8\sigma$	0.0476	Shaanxi, China, 1556 – killed 830,000 persons
Calamity Type 2 and higher	$1M \leq F$	$\mu + 10.8\sigma \leq F < \mu + 110.8\sigma$	0.0023	No historical records

5.2. Severity classification based on block maxima method using location-based blocks

The approximate value of the mean and standard deviation of the data set produced by the location-based block maxima method is 28,000 and 90,000, respectively. The fatality severity levels from 'Emergency' to 'Calamity Type 2 and higher' are represented in Table 14.

The probability of a lower level disaster (Emergency to Catastrophe Type 1) being the extreme disaster is higher for the location-based data set ($0.77 > 0.22$) compared to the event-based, worldwide data set; the probability of a higher level disaster (Catastrophe Type 2 and above) being the extreme disaster is lower. This result is to be expected because a single country, even over the full time period, is less likely to have a massive disaster in comparison with the world over a large number of extreme events, in this case 100.

6. Discussion

In this study, two different fatality-based severity levels were introduced based on three different methods of determining extreme events. Because each method has advantages and disadvantages, it is difficult to choose the best one. Each method creates its own extreme events sample and, thus, a different extreme value distribution. However, to have a better understanding of each fatality-based severity level, the strengths and weaknesses of each method need to be investigated.

*R*th order and block maxima event-based methods lead to similar results because the best data set for *R*th order was a 100 event block length with $R = 1$, which is the same as the set produced using block maxima method with 100 blocks. The merit of the 'peak over threshold' (POT) method compared with other extreme value methods is that it considers all the severe events that exceed a predefined level, which means that this method does not miss any extreme events over the threshold. However, the threshold method does not give us any information regarding events below the threshold level. Hence, we do not have any severity intervals for the events below the threshold level. In addition, different choices of threshold levels change the results significantly. Utilizing

Table 15. Dispersion of the extreme events for different methods.

Method	$1 \leq F < 30,000$	$30,000 \leq F$
POT	0%	100%
Event-based block maxima (same as *R*th order)	51%	49%
Location-based block maxima	80%	20%

Table 16. MAE values for different methods.

Method	PDF	MAE
POT	Pareto	0.019
Event-based block maxima (same as *R*th order)	FRECHET (3P)	0.015
Location-based block maxima	WEIBULL (3P)	0.024

the threshold method may lead to no values being chosen in some consecutive years, which means overlooking the most severe events in some intervals and incurring extra expenditures due to over-designing. While the threshold method may neglect the extreme events below the threshold, the block maxima method covers most of the extreme events both below and over the threshold as it picks the maximum as the extreme event in each block. However, modelling only block maxima implies that we omit a lot of data if a detailed recording of the studied phenomenon is available (Omey, Mallor, & Nualart, 2009). For instance, in this study, while 58 extreme events have more than 30,000 fatalities for the threshold method, there are only 29 for event-based and 23 for location-based extreme events with more than 30,000 fatalities. Therefore, 50% and 60% of the extreme events with more than 30,000 fatalities are omitted using event-based and location-based block maxima, respectively. More details are provided in Table 15.

However, extreme events are more and better dispersed using the block maxima method than with the threshold method as there are extreme events with fatalities fewer than 30,000. Moreover, comparing the MAE values of the three different methods, shown in Table 16, reveals that the block maxima method has a lower MAE value, which indicates that in block maxima, the extreme value distribution better fits the empirical data.

The classification method based on event-based block maxima is compared with the intensity scales (MMI, Rossi-Forel, Japanese, and European) according to the maximum number of fatalities and damaged houses recorded for each MMI level. The results indicated in Table 17 show that fatalities and damages begin at level IV on the intensity scales, which is equivalent to level I (Emergency) in the proposed method. This difference is reasonable because the proposed method is based on extreme events. Therefore,

Table 17. Comparison between proposed scale and intensity scales.

Fatality-based disaster scale	Fatality range	MMI scale	Rossi-Forel	Japanese	European
Emergency	$1 \leq F < 10$	II–IV	I–V	I–III	II–IV
Disaster Type 1	$10 \leq F < 100$	V	V–VI	III	V
Disaster Type 2	$100 \leq F < 1000$	VI	VI–VII	IV	VI
Catastrophe Type 1	$1000 \leq F < 10{,}000$	VII–VIII	VIII– to IX–	IV–V	VII–VIII
Catastrophe Type 2	$10{,}000 \leq F < 100{,}000$				
Calamity Type 1	$100{,}000 \leq F < 1M$	IX–XII	IX+ to X	V–VII	IX–XII
Calamity Type 2 and higher	$1M \leq F$	–	–	–	–

earthquakes that have a minimal effect on human lives are not included in proposed scale. Another outcome is the fact that all levels from 9 to 12 on the intensity scales are covered in level VI of the proposed scale.

7. Conclusion

The paper discusses a comprehensive statistical study about earthquake fatalities. The PDF of historical earthquake fatalities is investigated and the distribution of the extreme events is determined using three different approaches: block maxima (event-based and location-based), *R*th order, and the threshold method. The fatality-based severity level is introduced for earthquakes. It is shown that severity classification based on block maxima has more detailed severity classes and the lowest MAE; hence, it is superior to the other two methods for determining the severity level of extreme events.

For the location-based approach, the probability of a lower level extreme event (Emergency to Catastrophe Type 1) occurring is substantially higher than the same probability estimated via the event-based approach ($0.77 > 0.22$). Consequently, the probability of a higher level extreme earthquake is lower in the location-based approach. This result is to be expected since the probability that a massive disaster over 100 events occurs is higher when the data from the world is considered than the probability of such an extreme event occurring in a single country, even over the full time period.

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